

Building with Meaning: An Engineering Portfolio Rooted in Purposeful Design

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Position in Context: Bridging Dreams and Discipline

Before entering university, I saw engineering as a bold way to challenge limitations. Growing up in a rural village in China, where a woman's role was traditionally confined to the home, I turned to engineering not just as a career path, but as a form of resistance, a declaration that I could become whoever I wanted to be. I watched the women around me perform the same domestic chores every day, expected to follow a script that had been written for them before they had a chance to write their own. To me, that kind of life felt constrained, repetitive, and predefined. I wanted more than that. I wanted to create, to explore, and most importantly, to help people in a way that felt truly meaningful.

A week before university started, I was still holding onto those big ideas, eager, hopeful, and ready to make an impact. That feeling deepened during Frosh Week, when I learned the story behind the iron ring. I remember listening to an upper-year student speak about how the ring was a reminder of both pride and humility, how it brushed against the paper every time you signed your name as an engineer. That image stayed with me. It made engineering feel real, not just technical, but human, grounded in responsibility to others. I walked home that day filled with excitement, telling my family that I had finally found my purpose. It wasn't just about escaping old expectations anymore; it was about stepping into a profession rooted in service, care.

Through my engineering courses, I realized my view of engineering, as an escape from a boring, predefined life, was only part of the story. I thought passion and the desire to help were enough, as if choosing this path alone would lead to meaningful impact. One turning point came during my very first CIV lecture, when we were introduced to one of the three fundamentals of engineering: "Before solving the question, you must know the answer." At first, it sounded like a paradox, but what it really meant was that engineering isn't about blindly chasing ideas. It's about precision. It's about thinking critically, building intuition, and being able to predict outcomes before committing to a solution. That lesson shifted my perspective: I realized that meaningful impact in engineering doesn't come from inspiration alone, it comes from a balance between the desire to help and the discipline to design thoughtfully.

This realization made me reflect on the position I had shared at the beginning of the year. In my original position statement, I talked about dreaming big, but I've since learned that dreaming alone isn't enough. My ideas were abstract, lacking concrete experience and practical grounding. What I didn't yet understand was how to bring those values to life through design. Now, at the end of my first year in Engineering Science, I'm starting to close the gap between vision and execution. Through projects like the CIV bridge, Praxis I, and Praxis II, I've come to see engineering design as a balance between creative inspiration and structured logic. I'm learning how to build with both purpose and precision. As I move into more advanced design work, I carry this mindset forward: engineering isn't just my way out, it's how I turn ideas into impact, by balancing the desire to help with the discipline to build thoughtfully.

This portfolio captures my journey as a first-year Engineering Science student, documenting the major design projects I've worked on throughout the year. Each section highlights not only the outcomes of these projects, but also the processes, challenges, and decisions that shaped them. I reflect on the concepts, tools, models, and frameworks (CTMFs) that guided my design work, from early-stage needfinding and brainstorming to technical analysis, prototyping, and presentation.

Project I – CIV102 Matboard Bridge

Project Overview

The CIV102 Matboard Bridge Design Project challenged teams to design, analyze, and construct a small-scale structural bridge using only one sheet of matboard ($813\text{ mm} \times 1016\text{ mm} \times 1.27\text{ mm}$) and two tubes of contact cement. Working in teams of three or four, students were tasked with designing the central span of a three-span box girder bridge, which would support a moving train load and be subjected to increasing weights until structural failure occurred. The primary goal was to create the strongest and most weight-efficient bridge possible, while making accurate predictions about the failure load based on sound engineering principles.

The bridge had to span 1200 mm between supports, with total bridge length ranging from 1250 mm to 1270 mm. Teams had to ensure the bridge complied with dimensional and loading constraints, including maintaining a flat, level bridge deck and accommodating the passage of a 400 N train. If the bridge survived the first load pass (Load Case 1), it would be subjected to progressively increasing train weights (Load Case 2), eventually leading to failure. The project emphasized both structural performance and prediction accuracy.

Design and analysis were grounded in concepts from beam theory and thin plate buckling theory. Students calculated internal forces and moments by constructing Shear Force Diagrams (SFDs) and Bending Moment Diagrams (BMDs) for different train positions along the span. Using these diagrams, they created stress envelopes, which represented the maximum internal forces at each location across all load cases. With this data, teams determined applied stresses (tensile, compressive, and shear) and compared them to material strength limits and buckling capacities, calculating Factors of Safety (FOS) for each failure mode.

Structural geometry was defined by cross-sectional parameters such as flange width, height, and thickness, which were iteratively adjusted using hand calculations and custom MATLAB or Python scripts. Teams also had to address practical challenges not captured by theoretical models, such as splice connections (due to the limited length of the matboard), local shear and bearing failure near supports, and loss of cross-sectional rigidity. To reinforce the bridge and control deformation, vertical diaphragms were installed throughout the structure.

Deliverables included a comprehensive design report, engineering drawings, a detailed time log, and complete design calculations, both by hand and via code. The final product, the physical bridge, was tested in a lab setting and evaluated based on its maximum load before failure, prediction accuracy, and strength-to-weight ratio. This project challenged teams to practice real-world structural design, integrate theoretical concepts with hands-on prototyping, and develop a deeper appreciation for the complexity, trade offs, and responsibilities involved in civil engineering.

CTMF #1: Risk Management Framework

Purpose: To identify, assess, and manage uncertainty in a project so that one can make better decisions, reduce failure, and design responsibly.

Risk Management Framework Overview:

1. Objectives

- Every project has goals
- These goals come with uncertainty -> where risk starts
- Ask: what could go wrong while trying to meet these objectives?

2. Risk

- Risk = Likelihood x Impact
- Likelihood = How probable is it? (1-5 scale)
- Impact = How bad is it if it happens? (1-5 scale)
- List down the possible risks (eg. material failure to bridge, wrong calculation, etc.)

Impact Scale

Impact Level	Description
1-Negligible	Minimal effect, no significant consequences
2- Minor	Slight effect, minor delays or adjustments
3-Moderate	Noticeable impact, design needs revision
4-Significant	Serious effect, may cause partial failure
5-Servere	Severe impact, complete failure

Likelihood Scale

Impact Level	Description
1-Very Unlikely	Extremely unlikely, would need unusual circumstances
2- Unlikely	Not expected, but possible
3-Possible	Could happen, had occurred before
4-Likely	Expected to happen under common conditions
5-Very unlikely	Expected in most cases

Risk - effect of uncertainty on objectives/activities

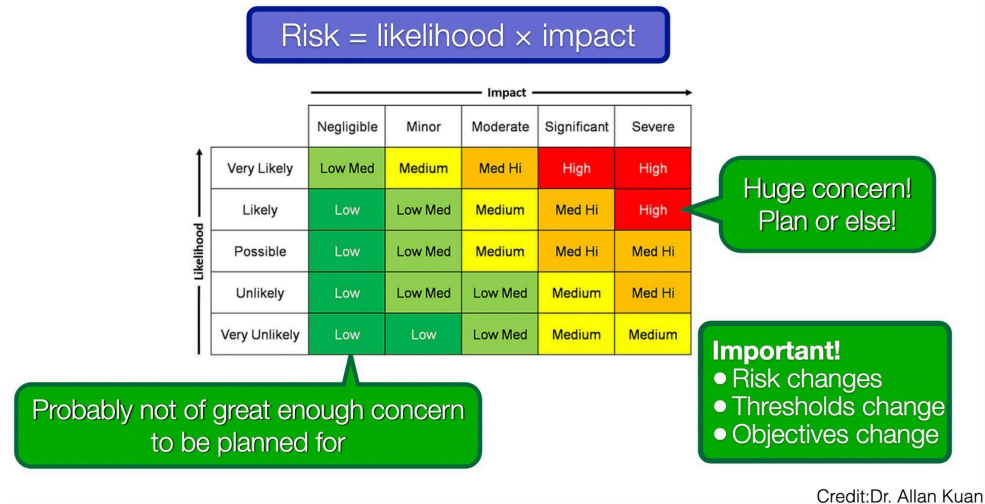


Figure 1. Risk calculation diagram.

3. Risk Treatment

- Mitigation: reduce the change or severity
- Avoidance: remove the risk entirely
- Transfer: shift risk to something else
- Acceptance: live with it if it's low enough risk

4. Residual Risk

- New risk level after treatment
- New risk = New Likelihood × New Impact
- Ask: Did our plan actually reduce the risk?

5. Risk Thresholds

- Compare residual risk to what your team is ok with
- Ask: is this acceptable based on team values and goals?

6. Decision Making

- Use everything above to inform your final design decisions
- Modify plans if risk is too high or if better trade-offs are found



Figure 2. Over structure of risk management framework.

My teammates for the CIV bridge project were Harley, Sasha, and Ahmad. Early on, as a team, we began by trying to decide what type of bridge we wanted to design. Initially, we spent time casually searching for “cool bridge designs,” but this approach lacked direction and didn’t lead to meaningful progress. It wasn’t until we revisited the assignment requirements more carefully that we shifted our approach. From there, we began considering factors such as material limitations, load paths, and failure risks, an approach that, while not formally planned, resembled a basic form of risk management.

From our objectives, we wanted a bridge that can: use the least amount of material, carry the most weight. Thus, using the framework looked like this:

1. objectives
 - Use the least amount of material
 - Carries the most weight
2. Risk

Risk	Likelihood	Impact
Beam buckling under compression	3	5
Splice joint failure	4	4
Underestimating stress distribution	3	4
Using too much material(overbuild)	4	2

3. Risk Treatment

- Buckling: increase flange height, add diaphragms
- Splices: overlap and reinforce double layers
- Stress analysis: use MATLAB/Python & stress envelopes to visualize critical areas
- Material use: iterate cross-section sizes to avoid overdesigning

4. Residual Risk

- Not applicable to us since this was used to decide what design to use

5. Risk Thresholds

- Trade off decision: “We’d rather risk overdesigning slightly than have the bridge collapse early.”

We used the Risk Management Framework during the framing stage to help us interpret our objectives and explore possible design directions. The goal was to design a bridge that used minimal material while still carrying a high load, but we quickly realized that not all design options carried equal levels of risk. The most useful part of the framework at this stage was the focus on identifying and assessing risks early on. It helped us move beyond just picking designs that looked strong or interesting, and instead think critically about which ones made sense given our constraints. We used the concept of likelihood and impact, even informally, to compare truss, suspension, cantilever, and beam bridges. The high-risk options (due to complexity or instability) were eliminated early, which helped us narrow our framing around the beam bridge. Although we didn’t use a formal risk matrix or assign exact numbers, this early filtering process was valuable.

Note that using the Risk Management Framework helped us do more than just eliminate high-risk options, it also revealed what we valued as a team. This CTMF is helpful in scoping design space, as well as establishing team values. One of the key trade-offs we agreed on was that we’d rather risk overdesigning slightly than have the bridge collapse early. That choice guided us toward the beam bridge, not because it was the most optimized, but because it gave us control, predictability, and room to design with confidence. Looking back, that decision felt right. Even in this early framing stage, we weren’t just aiming to make something that worked, we were thinking about how and why it worked. That mindset shaped how we framed the project from the beginning: a balance between doing something bold and doing it responsibly.

Overall, I would recommend using the Risk Management Framework during the framing stage if your goal is to narrow down your design space and make decisions based on constraints, trade-offs, and feasibility. It’s a helpful tool when you’re trying to filter out high-risk directions early on and commit to a grounded starting point. However, I wouldn’t recommend relying on it too heavily if your goal at that stage is simply to generate as many ideas as possible. Risk management is best used when you’re ready to bring focus, not when you’re still exploring the full creative range of possibilities. Like most tools, its value depends on when and how it’s used.

CTMF #2: SCAMPER

Purpose: To explore variations and alternatives of an existing design by applying seven thinking prompts. Helps expand your design space and think beyond obvious solutions.

SCAMPER stands for:

1. S - Substitute
 - Replace a material, component, or process with another
 - Ask: What can I replace?
 - Useful for improving performance, reducing cost, or using available resources
2. C - Combine
 - Combine two functions, shapes, or systems into one
 - Ask: What can I merge or blend?
 - Great for reducing complexity and saving space/material
3. A - Adapt
 - Adapt a solution or idea from another structure or field
 - Ask: What else is like this? Can I borrow from another context?
 - Great for promoting cross-disciplinary thinking and innovation
4. M - Modify
 - Modify thickness, span, cross-section shape, etc
 - Ask: What can I change in size, shape, angle, or configuration?
 - Helps fine-tune performance characteristics
5. P - Put to Other Use
 - Find a new purpose or configuration for a part
 - Ask: Can this be used differently?
 - Encourages efficient material use and multifunctionality
6. E - Eliminate
 - Remove non-essential components, reduce redundancy
 - Ask: What can I remove or simplify?
 - Supports simplicity and lightweight design
7. R - Rearrange/ Reverse
 - Flip, rotate, or reorganize components
 - Ask: Can I change the order or orientation?
 - Useful for uncovering overlooked configurations

During the diverging stage of our CIV project, once we had narrowed our framing to a beam bridge, we used SCAMPER(not consciously) as a way to explore a range of possible design directions. We didn't settle on one beam configuration immediately. Instead, we generated many design variations by tweaking features, combining ideas, and removing nonessential elements. This helped us expand our idea space before converging on the final design.

We made heavy use of the Modify principle through iterative dimensional changes. Using Figma, we constantly adjusted the proportions of key components, such as web height, flange width, and internal bracing placement, to test their effect on performance and constructability. These sketches allowed us to rapidly visualize new versions of our design without physically building anything, which saved time and encouraged experimentation. Each member contributed feedback and suggestions, allowing us to build on each other's ideas in real time.

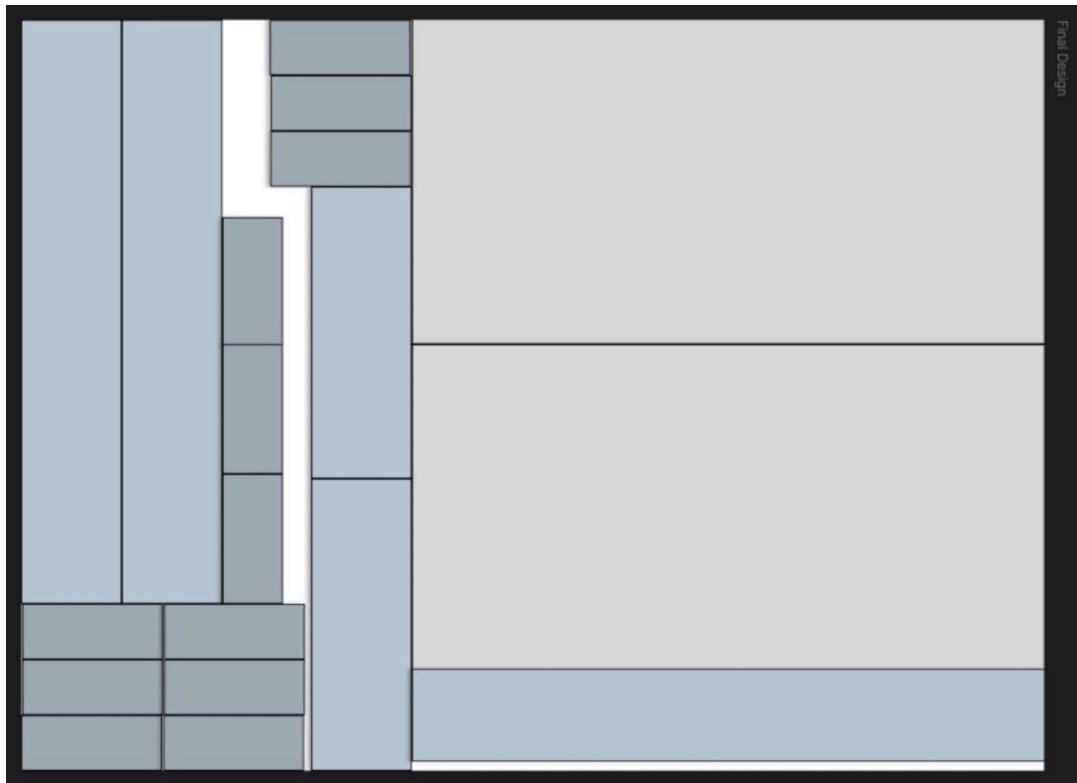
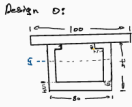


Figure 3. Final bridge dimension using Figma.

Substitute and Eliminate also appeared frequently in our process. For example, we considered replacing a solid internal structure with double webs or using partial flanges instead of full-length ones to save material. In some cases, we questioned whether certain internal supports were necessary at all, leading us to remove or reduce them when they didn't contribute meaningfully to structural strength. This kind of elimination was supported by data from our hand and coded calculations.

FOS against failure - design 0
 Design 01



From SFE and BME:
 Maximum moment, $M(1) = 6.9 \times 10^3 \text{ N}\cdot\text{m}$
 Maximum shear force, $V(1) = 22.9 \text{ kN}$

Geometric Properties:

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} = \frac{h_2 w_2 (u_2 + \frac{h_2}{2}) + h_2 w_2 (u_2) + 2 h_2 w_2 (\frac{u_2}{2}) + 2 w_2 z_2 (u_2 - \frac{h_2}{2})}{h_2 w_2 + h_2 w_2 + 2 h_2 w_2 + 2 w_2 z_2} = 41.43 \text{ mm}$$

$$I = \sum C_{xi} + A_i \bar{y}_i^2 = \frac{w_2 h_2^3}{12} + w_2 h_2 (u_2 + \frac{h_2}{2} - \bar{y})^2 + \frac{w_2 h_2^3}{12} + w_2 h_2 (\bar{y} - \frac{h_2}{2})^2 + 2 (\frac{w_2 h_2^3}{12} + w_2 h_2 (\bar{y} - \frac{h_2}{2})^2) + 2 (2 w_2 z_2^3 + w_2 z_2 (h_2 - \frac{h_2}{2} - \bar{y})^2)$$

$$= 0.915 \times 10^8 \text{ mm}^4$$

$$Q_{max} = E(A_i \bar{y}_i^2) = (w_2 h_2) (\bar{y} - \frac{h_2}{2})^2 + 2 (z_2 w_2) (\bar{y})^2 = 43.60 \text{ mm}^3$$

$$Q_{she} = E(A_i \bar{y}_i^2) = h_2 w_2 (\bar{y} - \frac{h_2}{2})^2 = 2210 \text{ mm}^3$$

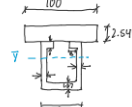
Failure Modes:
 Compression: $\sigma_{comp} = \frac{M(\bar{y})}{I} = 6.76 \text{ MPa}$
 Tension: $\sigma_{tens} = \frac{M(\bar{y})}{I} = 6.86 \text{ MPa}$
 Shear: $\tau_{she} = \frac{Q_{max} V_{max}}{2 I b} = 1.05 \text{ MPa}$
 Torsion: $\tau_{tors} = \frac{Q_{she} V_{max}}{2 I b} = 0.103 \text{ MPa}$

Thin plate buckling:
 Case 1: $\sigma_{comp} = \frac{4 \pi^2 E}{12(1-\nu^2)} \left(\frac{h_2}{w_2 - 2 z_2} \right)^2 = 4.61 \text{ MPa}$
 Case 2: $\sigma_{comp} = \frac{0.4 \pi^2 E}{12(1-\nu^2)} \left(\frac{h_2}{(u_2 - h_2)/2} \right)^2 = 23.5 \text{ MPa}$
 Case 3: $\sigma_{comp} = \frac{6 \pi^2 E}{12(1-\nu^2)} \left(\frac{u_2}{w_2} \right)^2 = 34.4 \text{ MPa}$
 Case 4: $\tau_{tors} = \frac{5 \pi^2 E}{12(1-\nu^2)} \left(\frac{u_2}{w_2} \right)^2 \left(\frac{w_2}{u_2} \right) = 6.22 \text{ MPa}$

FOS's:
 Factor of safety = $\frac{\sigma_{tens,allow}}{\sigma_{tens}} = 4.38$
 Factor of safety = $\frac{\sigma_{comp,allow}}{\sigma_{comp}} = 1.08$
 Factor of safety = $\frac{\tau_{she,allow}}{\tau_{she}} = 3.31$
 Factor of safety = $\frac{\tau_{tors,allow}}{\tau_{tors}} = 13.13$
 Factor of safety = $\frac{V_{max,allow}}{V_{max}} = 5.01$
 Factor of safety = $\frac{M_{max,allow}}{M_{max}} = 0.782 \rightarrow$ $\frac{\sigma_{comp,allow}}{\sigma_{comp}} = 0.782$ $\rightarrow$ $\frac{V_{max,allow}}{V_{max}} = 0.782$ $\rightarrow$ $\frac{M_{max,allow}}{M_{max}} = 0.782$

Figure 4. Hand Calculations for the Factors of Safety of Design 0 Under Load Case 1.

FOS of Final Design Against Load Case 2
 Monday, November 25, 2024 8:57 PM



From SFE and BME:
 Maximum moment, $M_{max} = 30.4 \text{ kN}\cdot\text{m}$
 Maximum shear force, $V_{max} = 75.1 \times 10^3 \text{ N}\cdot\text{m}$

Geometric Properties:

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} = \frac{h_2 w_2 (u_2 + \frac{h_2}{2}) + h_2 w_2 (u_2) + 2 h_2 w_2 (\frac{u_2}{2}) + 2 w_2 z_2 (u_2 - \frac{h_2}{2})}{h_2 w_2 + h_2 w_2 + 2 h_2 w_2 + 2 w_2 z_2} = 66.6 \text{ mm}$$

$$I = 2(I_{xc} + A_i \bar{y}_i^2) = \frac{w_2 h_2^3}{12} + w_2 h_2 (u_2 + \frac{h_2}{2} - \bar{y})^2 + \frac{w_2 h_2^3}{12} + w_2 h_2 (\bar{y} - \frac{h_2}{2})^2 + 2 (\frac{w_2 h_2^3}{12} + w_2 h_2 (\bar{y} - \frac{h_2}{2})^2) + 2 (2 w_2 z_2^3 + w_2 z_2 (h_2 - \frac{h_2}{2} - \bar{y})^2)$$

$$= 1.930 \times 10^8 \text{ mm}^4$$

$$Q_{max} = 2(A_i \bar{y}_i^2) = w_2 h_2 (\bar{y} - \frac{h_2}{2})^2 + 2 z_2 w_2 (\bar{y})^2 = 8.90 \times 10^3 \text{ mm}^3$$

$$Q_{she} = 2(A_i \bar{y}_i^2) = w_2 h_2 (\bar{y} - \frac{h_2}{2})^2 = 6.53 \times 10^3 \text{ mm}^3$$

Failure Modes:
 Compression: $\sigma_{comp} = \frac{M(\bar{y})}{I} = 2.06 \text{ MPa}$
 Tension: $\sigma_{tens} = \frac{M(\bar{y})}{I} = 3.59 \text{ MPa}$
 Shear: $\tau_{she} = \frac{Q_{max} V_{max}}{2 I b} = 0.552 \text{ MPa}$
 Torsion: $\tau_{tors} = \frac{Q_{she} V_{max}}{2 I b} = 0.06 \text{ MPa}$

Thin plate buckling:
 Case 1: $\sigma_{comp} = \frac{4 \pi^2 E}{12(1-\nu^2)} \left(\frac{h_2}{w_2 - 2 z_2} \right)^2 = 8.94 \text{ MPa}$
 Case 2: $\sigma_{comp} = \frac{0.4 \pi^2 E}{12(1-\nu^2)} \left(\frac{h_2}{(u_2 - h_2)/2} \right)^2 = 23.49 \text{ MPa}$
 Case 3: $\sigma_{comp} = \frac{6 \pi^2 E}{12(1-\nu^2)} \left(\frac{u_2}{w_2} \right)^2 = 34.4 \text{ MPa}$
 Case 4: $\tau_{tors} = \frac{5 \pi^2 E}{12(1-\nu^2)} \left(\frac{u_2}{w_2} \right)^2 \left(\frac{w_2}{u_2} \right) = 2.13 \text{ MPa}$

FOS's:
 Factor of safety = $\frac{\sigma_{tens,allow}}{\sigma_{tens}} = 2.88$
 Factor of safety = $\frac{\sigma_{comp,allow}}{\sigma_{comp}} = 8.37$
 Factor of safety = $\frac{\tau_{she,allow}}{\tau_{she}} = 7.25$
 Factor of safety = $\frac{\tau_{tors,allow}}{\tau_{tors}} = 24.4$
 Factor of safety = $\frac{V_{max,allow}}{V_{max}} = 2.88$
 Factor of safety = $\frac{M_{max,allow}}{M_{max}} = 4.25$

Figure 5. Hand Calculations for the Factors of Safety of Design 0 Under Load Case 2.

The Rearrange aspect of SCAMPER was applied when experimenting with diaphragm placement. Instead of placing them at uniform intervals, we discussed aligning them with key high-stress zones identified in our Shear Force and Bending Moment Envelopes. This subtle shift allowed us to use fewer diaphragms more efficiently, reducing weight while maintaining performance.

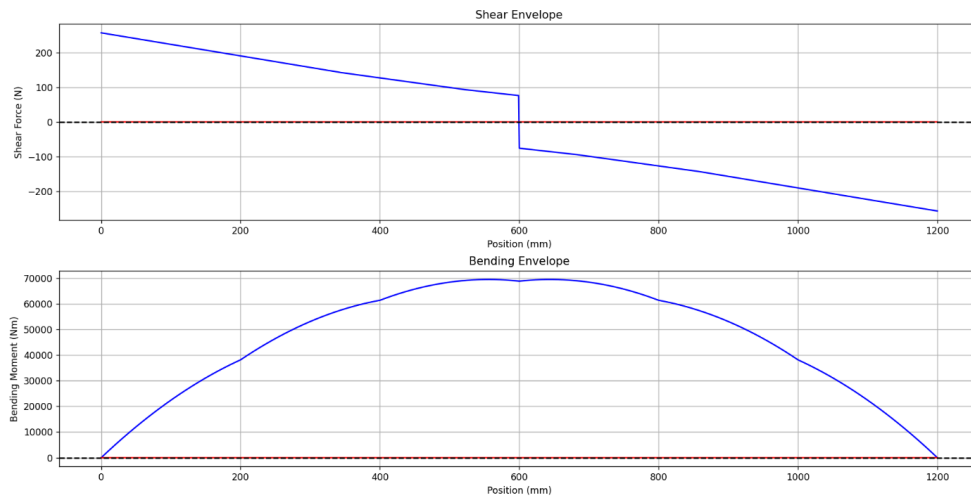


Figure 6. Shear Force and Bending Moment Envelopes using Python.



Figure 7. Developing Python script for Shear Force Envelope and Bending Moment Envelope.

SCAMPER turned out to be one of the most helpful tools during the diverging stage of the CIV project. Instead of jumping from idea to idea blindly, we had a method to explore variations in a focused way. That kind of thinking, wide exploration grounded in logic, ended up shaping a lot of our design process.

One of the things I appreciated most about SCAMPER was how flexible it was. We didn't feel the need to apply every part of it equally. Some prompts, like "put to other use," didn't offer much in our context, and so we just moved on. Others, like "modify" and "eliminate," were extremely effective, especially when we used Figma to change dimensions and simplify sections of the design. The framework gave us

structure when we needed it, but it never boxed us in. It expanded our design space without pushing us too far off course, exactly the kind of balance we needed at that stage.

I would absolutely use SCAMPER again. It fits naturally into the diverging stage of the design process, but it can also support framing by helping teams challenge early assumptions. It's especially valuable when you want to keep creativity alive without losing sight of practicality. SCAMPER can open up so many possibilities that, without a system for filtering or prioritizing ideas, it can quickly become overwhelming. In group settings, it also needs structure, otherwise, ideas pile up without moving forward. Pairing it with a Pugh chart or clear criteria would make it even more powerful.

Looking back, the value of SCAMPER was less about generating something new out of nowhere, and more about making what we already had sharper, more intentional, and better understood. It gave us a way to think through possibility.

CTMF #3: Gap Analysis

Purpose: to identify the difference (or “gap”) between a current design and the desired goals, performance targets, or user needs.

How to use it:

NOTE. This is just a step guide that I found helpful myself, but there are no actual steps to gap analysis.

1. Set your targets
 - Define your design requirements or goals
 - These become your benchmark
2. Evaluate current designs or concepts
 - Choose the designs you’re comparing
 - Collect performance data(test results, hand calculations, simulations, etc)
3. Identify the “gap”
 - Compare current performance to the target
 - Ask: What is this design missing? Where does it fall short?
4. Analyze and prioritize
 - Ask: Which are most critical to fix?
 - Choose which design has the smallest or most manageable gap
5. Decide what actions to take
 - Adjust the design to close the gap

We used gap analysis to pinpoint the most critical weakness in our design. One of the clearest gaps we identified was related to bending stress at mid-span, our initial structure was too shallow, and calculations showed that the stress exceeded safe limits. It didn’t take long to realize that increasing the height of the bridge would improve the moment of inertia and reduce bending, bringing the stress back within acceptable bounds. The change wasn’t dramatic, it was a relatively small adjustment, but it had a big impact on performance.



Figure 8. Notice the bridge is thinner and taller than a normal beam bridge.

What made gap analysis useful was that it helped focus our thinking. It filtered out distractions and brought us directly to the change that would matter most. Looking back, that was a valuable takeaway for me: not every design decision has to feel exciting or bold. Sometimes the most effective changes are subtle, just a small refinement can bring everything into better alignment. That kind of thinking has become more natural to me over time and how I view engineering design overall: less about chasing big ideas for myself, and more about designing with clarity, purpose, and care.

Gap analysis helped us assess our direction against measurable goals and close the distance between “almost working” and “working well.” However, gap analysis isn’t always necessary, or even helpful, for everyone. For me personally, it felt very intuitive. I realized I had already been thinking this way without knowing it was a formal CTMF. For people who feel stuck or aren’t sure where a design is falling short, gap analysis can be incredibly helpful. It brings you right to the point. But for those who already approach design with a diagnostic mindset, it might feel redundant. In that case, I’d recommend using another tool, perhaps a tool that pushes for creativity or reframing instead.

In short, I would recommend gap analysis when there’s a clear problem to solve or a target to meet. It’s a great tool for grounding decisions and moving forward with intention. But it’s not the only path forward, and sometimes, your own instincts will lead you there first (“Before solving the question, you must know the answer!”).

Project II – Praxis I: Reducing Cat Litter Stench

Project Overview

The Praxis I design project challenged student teams to explore a real-world problem through a structured engineering design process. Rather than solving a clearly defined task, teams were given a broad design brief and were expected to frame the problem themselves by identifying stakeholder needs, constraints, and opportunities. The design process emphasized four key phases: framing the problem, diverging to explore a range of possible solutions, converging through analysis and testing to identify the strongest concept, and finally presenting a credible, verifiable design recommendation based on evidence.

Our team chose to focus on the issue of reducing the stench associated with cat litter boxes, a problem that many cat owners experience but often struggle to address effectively. Initial research and stakeholder conversations revealed that litter tracking, where fine litter particles stick to a cat's paws and spread through the home, was a significant contributor to lingering odor. This insight allowed us to define a design direction focused not only on the box itself, but on the mechanisms by which odor spreads into the surrounding environment. We developed a set of design objectives that prioritized odor control, low maintenance, safety, and compatibility with existing litter box use.

Over the course of the project, we explored and tested several design concepts that addressed these objectives in different ways. Our ideas ranged from mechanical add-ons to passive modifications in form and material. Based on performance and alignment with user needs, we selected and refined the most promising option.

The final recommendation was a larger litter substrate, which significantly reduced litter tracking and, by extension, odor spread. This solution required no additional mechanical components, introduced no usability issues for the cat or the owner, and could be implemented with minimal disruption. It met all of our original objectives while maintaining simplicity and practicality, offering an efficient and scalable way to improve the everyday experience of cat ownership.



Figure 9. Praxis I group members.

CTMF #4: Stakeholder Analysis

Purpose: To help identify, understand, and prioritize the needs, concerns, and values of the people affected or involved in your design.

A Stakeholder is Anyone Who is:

- Directly using the product or system(eg. cat owner)
- Indirectly affected(eg. the cat, people living in the same house)
- Involved in maintaining or approving the design(eg. manufacturers, landlords)

Steps to Conduct a Stakeholder Analysis:

NOTE. This is just a step guide that I found helpful myself, but there are no actual steps to identifying stakeholders.

1. Identify all stakeholders
 - Think broadly: users, maintainers, decision-makers, etc.
2. Map their relationship to the problem
 - Who experiences the problem?
 - Who interacts with the system?
 - Who is responsible for upkeep?
3. Determine their needs, goals, and values
 - What are they trying to achieve?
 - What are their frustrations or fears?
4. Identify conflicts or trade-offs
 - One stakeholder's benefit might be another's inconvenience
 - Eg.reducing odor (owner) VS. comfort (cat)
5. Prioritize stakeholder importance
 - Direct users often carry more weight than peripheral actors

Stakeholder Overview

Stakeholder	Role	Needs/Values	Conflicts or Trade-Offs	Influence on Design
Cat Owner (Primary)	Direct user, experiences odor	Reduce stench, easy to clean, minimal maintenance, affordable	May conflict with design complexity	Highest : main decision-maker and evaluator
Cat (Secondary)	End user, interacts physically with product	Comfort, safety, easy access, behavioral	Odor control may require modifications that	High: if the cat doesn't accept the design, it fails

		familiarity	feel unfamiliar	
Other Residence	Passive users, live in the same space	Reduced ambient odor, no litter mess	May expect benefits without maintaining the box	Medium: they influence perceived success of solution
Retailers/Suppliers	Distribution, source of litter or products	Product availability, demand, ease of shelf placement	Uncommon substrate sizes may limit retail viability	Low: indirect influence via supply chain
Manufacturers	Produce substrates or litter box components	Scalable production, material cost, product compatibility	May resist change due to retooling or low demand	Medium: may constrain what designs are feasible to produce

We used stakeholder analysis early in our project as part of the framing stage. At the time, it was a very helpful tool, not just for identifying who was involved, but as a new Praxis student, to really think about what roles people play in shaping a design. Using a structured chart to distinguish between primary, secondary stakeholders gave me clarity on who the design was really for, who was affected, and who had indirect influence. It also helped organize our priorities: what trade-offs we could make, and what constraints we couldn't ignore.

At that early stage, the most useful aspect was simply listing out the stakeholders and writing down their needs. As someone still getting used to thinking like an engineer, I didn't naturally see the difference between, say, the person cleaning the litter box and the cat using it. The tool helped bridge that gap. It provided a foundation not just for this project, but for the rest of the design process. We used stakeholder needs to define design criteria, evaluate alternatives, and justify our final recommendation. In that way, stakeholder analysis wasn't just framing a design, it fed directly into how we diverged (by focusing on stakeholder-driven concepts), how we converged (by comparing which ideas best met those needs), and how we presented the final solution.

Now, with more experience, I can apply stakeholder analysis more naturally, thinking about users and needs without needing a guide. But even so, I wouldn't skip the tool entirely. There's value in going step by step, not because I can't guess what people need, but because engineering design requires clarity and discipline, not just intuition. That idea has shown up again and again in my experience: good design isn't always about being clever, it's about being thoughtful and getting the basics right. And for that, stakeholder analysis is one of the most fundamental and important tools.

I'd definitely recommend it to anyone starting out, especially for people new to knowing how to "think like an engineer". It's a strong framework for keeping the process grounded in real-world context. I wouldn't recommend skipping it even if you "think you know" what people want, because assumptions

tend to creep in without realizing. Stakeholder analysis really sets the direction of the entire project on solid ground.

CTMF #5: Classical Brainstorming

Purpose: To rapidly generate a large number of diverse ideas in a short time, without immediately judging or filtering them. The goal is to explore a wide idea space and promote creative thinking, especially during the diverging stage of the design process.

How it Works:

1. Set a clear goal or prompt
2. Follow core brainstorming rules
 - Defer judgement: no idea is dismissed or criticized
 - Go for quantity: more ideas = more creative potential
 - Encourage wild ideas: even strange ideas can spark usable ones
 - Build on the ideas of others; “yes, and...” instead of “no, but...”
3. Record everything
 - Use sticky notes, whiteboards, digital docs...
4. Cluster & reflect afterward
 - Group similar ideas, identify patterns, or mix ideas into new combinations

During the early stages of our cat litter box project, we used classical brainstorming in the diverging phase because it was fast, familiar, and easy to set up. It allowed everyone on the team to contribute freely and helped us quickly fill the board with ideas.

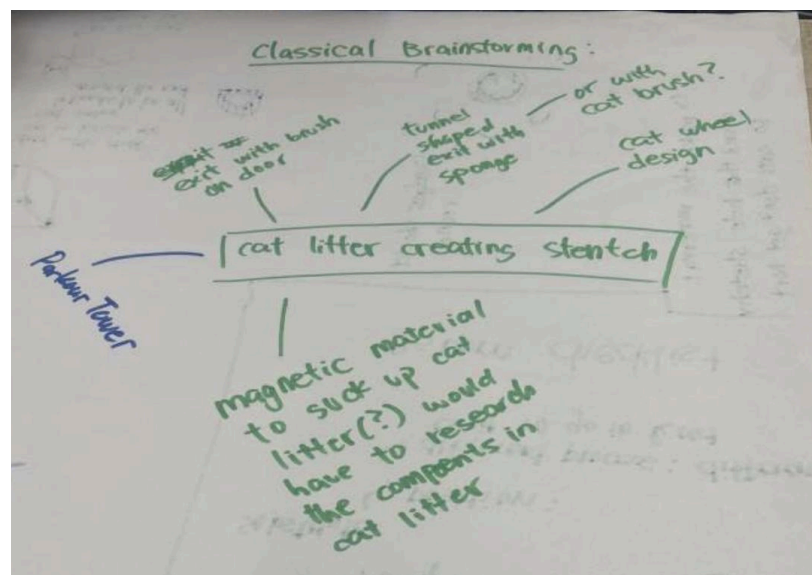


Figure 10. Classical brainstorming for designs to reduce cat litter stench. Some crazy ideas we had was eg. using magnetic materials.

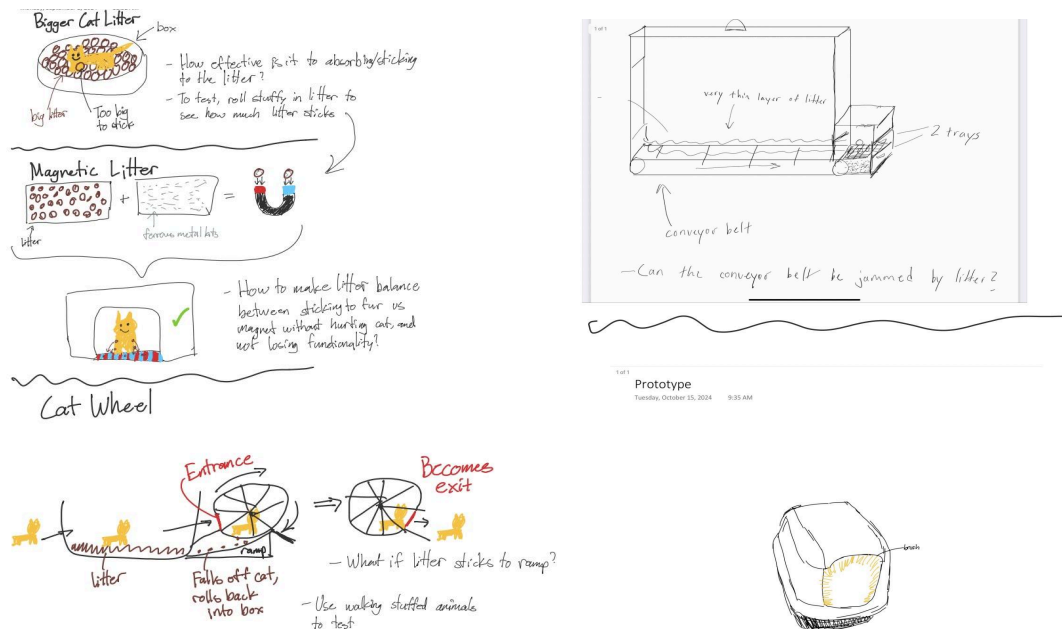


Figure 11. Drawn ideas from classical brainstorming.

But while it technically did what it's supposed to, expand our idea space, I've come to realize that it did so without a clear sense of direction. We were diverging, yes, but it was diverging for the sake of diverging, not because we were meaningfully pushing toward a better design outcome.

That's the core issue I found with classical brainstorming: it's often too open. It doesn't ask you to reflect on your design criteria, doesn't guide you toward structured exploration, and doesn't hold you accountable to your framing. And because it's so easy to do, it's just as easy to fall into using it by default. In our case, we stayed within the general topic of reducing cat stench, but many of the ideas lacked depth or relevance.

This experience made it clearer to me that every tool in the FDCR process needs to be used with purpose. Diverging should still be connected to what you're aiming for. Otherwise, you end up with volume, not value. I wouldn't necessarily say classical brainstorming is useless, but I would caution against relying on it too heavily, especially for teams who are still shaping their problem space. It risks making design feel shallow, like a list of random thoughts rather than a thoughtful exploration.

One of the most important lessons I've taken away from Praxis is that it's not enough to just apply design tools, you have to understand why you're using them. What purpose does this tool serve? Why choose this one over another? Tools aren't inherently valuable on their own, they only become meaningful when you apply them with intention. Intention and purpose are themes you'll see me return to throughout this portfolio. They've been central to how I've come to understand engineering design and continue to guide the way I approach each stage of the process.

CTMF #6: Morph Chart

Purpose: To break a design problem into sub-functions and systematically explore and combine different ways to fulfill each function. It helps diverge creatively or converge logically by mixing and matching partial solutions into full design concepts.

How to use it:

1. Identify key functions or components
 - What functions or features could you potentially add to your design?
2. List multiple options for each function
 - Each row = one function
 - Each column = possible solutions for that function
3. Form combinations
 - Pick one element from each row to create a complete design concept
 - You can create many unique combinations by rearranging options
4. Evaluate combinations
 - Use criteria
 - Filter out weak ideas, highlight promising ones
5. Select a leading concept
 - Choose the strongest combination for further development, testing, or prototyping
 - Optionally, refine or hybridize from multiple strong combinations

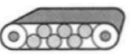
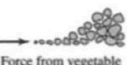
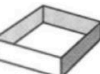
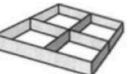
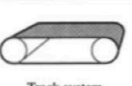
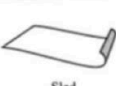
Ways of achieving the Feature/function				
Features/Functions of what you're trying to design	Option 1	Option 2	Option 3	Option 4
		 Triangular plow	 Tubular grabber	 Mechanical picker
	 Conveyor belt	 Rake	 Rotating mover	 Force from vegetable accumulation
	 Square mesh	 Water from well	 Slits in plow or carrier	
	 Packaging device	 Track system	 Sled	
	 Method of transportation	 Track system	 Sled	
	Power source	Hand pushed	Horse drawn	Wind blown

Figure 12. An in class example of a morph chart.

We used a morph chart to narrow down from broad concepts toward a more defined solution. While the tool is traditionally introduced as a way to diverge, by mixing and matching features to create new ideas, we actually found it most useful during convergence. At that point, we had already generated several partial concepts and wanted a clearer way to evaluate and combine the best parts of each. The morph chart helped us visualize what we had, organize it by function, and spot combinations that might not have come up through open discussion alone.

It was especially helpful following our classical brainstorming session, which, while useful for getting ideas flowing, left us with a scattered set of concepts. Brainstorming gave us quantity, but the morph chart gave us clarity. It allowed us to take the messy, unstructured ideas and sort them according to the key functional needs of our design.

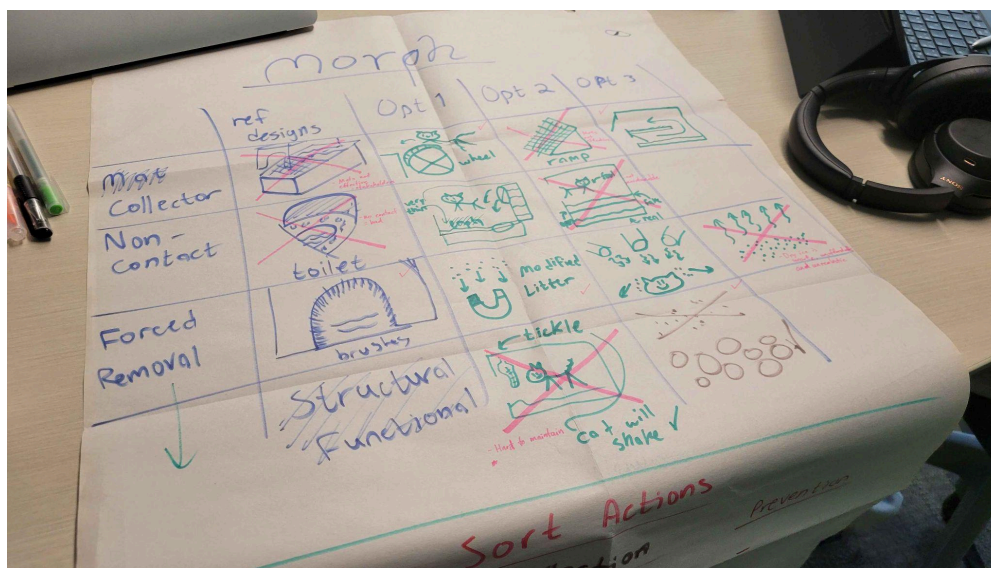


Figure 13. Certain functions being eliminated using a morph chart.

The most useful part of the chart was this forced structure. By listing out multiple options per function, it encouraged us to think modularly and spot strengths and weaknesses more easily. It gave us a clearer view of trade-offs and allowed us to justify decisions based on alignment with our objectives.

I wouldn't say there's anything about the morph chart I wouldn't use again, it's a very adaptable tool. But it only works well when your framing is strong and your criteria are clear. If used too early, or without intention, it can become just another random mixing exercise. Like any tool in the FDCR cycle, it only adds value when applied with purpose.

I would recommend the morph chart to teams who already have a pool of ideas and need a structured way to move forward. For us, it acted as a bridge between diverging and converging, transforming disorganized ideas into a focused, evaluative process. It brought structure after creative chaos, and helped us design with intention, not just inspiration.

Project III – Praxis II: Supporting Posture for Beginner Potters

My Praxis II team consisted of Rohan, Moervica, and Andy. Working alongside them, I experienced the most growth I've had in any team so far. I learned not only through the project, but through the way we challenged each other's thinking, supported one another, and approached design with honesty and care. I'm incredibly grateful to have met them. This team shaped not just the outcome of our project, but my entire perspective on what engineering design can and should be.

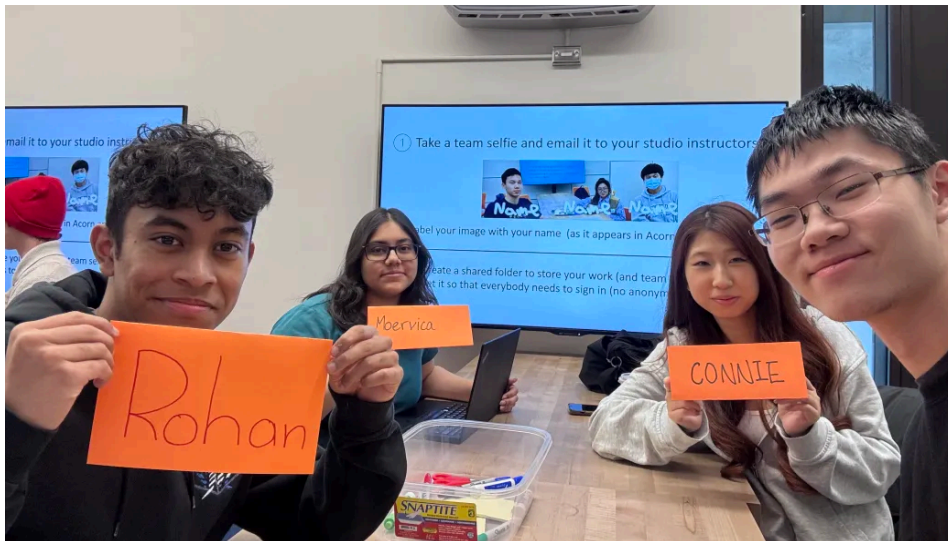


Figure 14. My first group photo with my Praxis II team.



Figure 15. Some fun times with my team members.

Project Overview

ERGOTHROW

Pottery Posture Corrector
with the MUD Clay Studio



The Opportunity: Posture Correction

The MUD Clay Studio is a pottery studio. Beginner potters in MUD often struggle to maintain proper posture during pottery, leading to discomfort and frustration.

Good Posture: keeping a straight back, leaning forward, and tucking your elbows into your hip.



Introducing the ErgoThrow

This Velcro-based design temporarily connects the elbow to the hip, guiding arm placement and promoting a natural forward lean. This ensures beginners keep good posture, while still allowing movement.

Prioritised Objectives:

- Reduce pain, guide posture
- Maintain freedom of motion
- Fit within the potter's workspace

The ErgoThrow temporarily restricts movement to guide posture, but the user can always detach the velcro fastening. Ensuring this freedom of movement adheres to the Studio's value of freedom of artistic expression.



2 of Our Main Design Decisions

1. The coloured buttons help **relate the size** the user wears the ErgoThrow at with **where the user has to put their elbows** on the belt, to promote adjustability and **inclusivity**, one of the key MUD values.
2. The belt was made **wider** so users didn't have to adjust the belt up and down to **ensure the arm-bands can reach** and stick to the band.



2 of Our Verification Tests

- **Posture**: graphing leaning angle over time to see **deviation from good posture** (aim: smallest standard deviation of angle)
- **Size**: parallel board test to measure **total size of the product** (aim: smaller product to ensure it fits in the workspace)

Results from Testing vs Other Designs

Posture: lowest standard deviation (3.09°)

Size: smallest (L×W×H = 23×18×7.5)



How to Use ErgoThrow

1. Put on the belt such that the velcro flap is on the front, and the edge buttons are on the bottom of the belt. Put on the arm-bands so they sit just above the elbow.
2. The button that the end of the flap is closest to correlates to the same-coloured buttons on the bottom of the belt. Place your elbows above these buttons.
3. Use the belt for the first few lessons until you begin to naturally keep this position.

CTMF #7: Challenge Assumptions

Purpose: To identify and question the hidden assumptions that limit creativity in problem-solving. Helps to break free of default thinking and explore new areas of the design space that would otherwise remain overlooked.

How to use it:

1. Start with your current understanding of the problem
 - These could be about the user, function, materials, environment, or what's "normal."
 - Ask: What are we assuming without realizing it?
2. List out those assumptions explicitly
 - Eg. posture correction must involve back support
3. Flip or challenge each one
 - Ask: What if this wasn't true?
 - Ask: What if we did the opposite?
 - Ask: Can we solve the problem without this assumption?
4. Use the challenged assumptions to redefine your design space
 - Organize your ideas into categories or directions inspired by what changed when the assumptions were questioned.
5. Generate new ideas based on those redefined spaces
 - This becomes a focused diverging phase with new, unconventional starting points.

We used Challenge Assumptions early in our Praxis II design process as a way to structure our diverging phase. At this stage, I had grown more mature in how I approached design tools. I no longer saw tools as boxes to check, I saw them as instruments that needed to be used with purpose.

By deliberately questioning those assumptions, we were able to redefine the boundaries of our design space. From there, we separated our exploration into three clear categories: body-based, wheel-based, and seat/stool-based solutions. This split didn't directly come from the tool itself, but it flowed naturally out of the conversations we had while challenging those initial assumptions. That's when I really began to understand what purposeful diverging meant, not just generating ideas, but exploring them with clarity, grounded in the problem we framed.

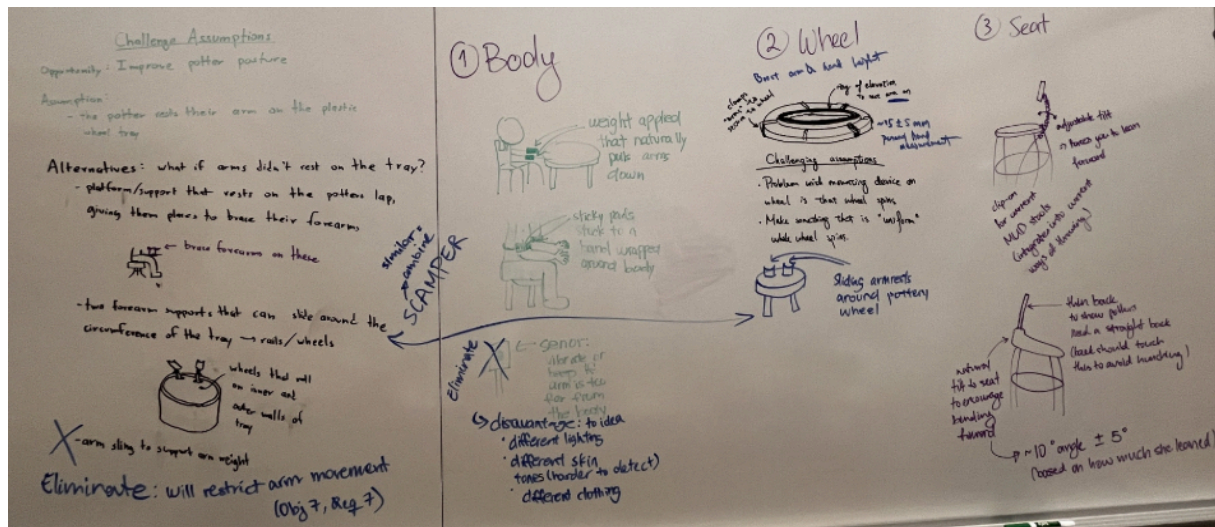


Figure 16. Using Challenge Assumptions to later on diverge to 3 main categories: body, wheel, seat.

Challenge Assumptions worked especially well at this point in the project because our framing was already strong. We knew what we were trying to solve, so this tool helped us push the edges of that space. In that way, it enabled us to diverge meaningfully, and set up the conditions for later convergence. While the tool wasn't used again in the later stages, the ideas it produced stayed with us and became inputs for tools like SCAMPER and attribute listing.

I would recommend Challenge Assumptions because it's a structured way to unlock creative, unconventional ideas. It's especially powerful when used with intention. For best results, I'd suggest pairing it with another tool or using it as a launchpad to diverge more purposefully into other directions. It works best when it's not just about idea generation, but about expanding your design space with clarity.

CTMF #8: Pairwise Comparison

Purpose: To objectively compare a set of options by evaluating them two at a time, based on defined criteria. It helps teams prioritize or rank alternatives when it's difficult to make a direct decision between multiple options.

How to use it:

- List all options you want to compare
 - These could be design concepts, features, materials, etc.
- Set up the comparison matrix
 - Place the same list of options on both the rows and columns of a table
- Compare each pair
 - For each cell in the matrix, choose which of the two options is stronger based on your criteria
- Assign a score
 - Give 1 point to the "winner" of each pair and 0 otherwise

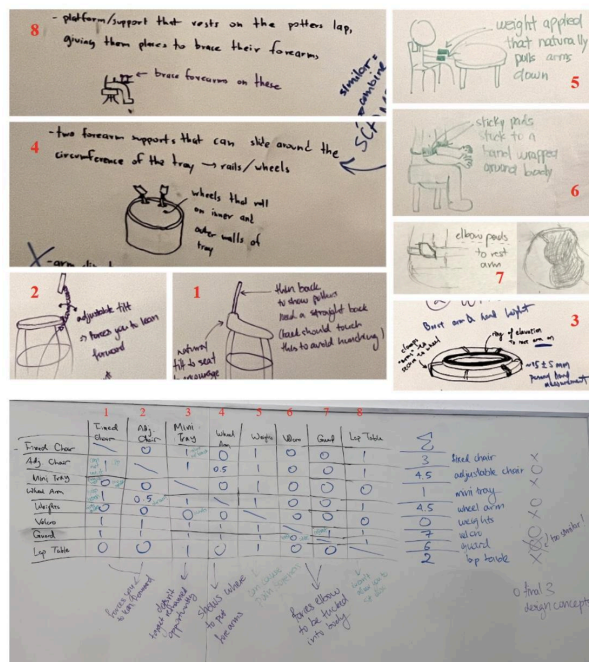


Figure 17. Pairwise comparison of concepts.

Converging with Community's Values

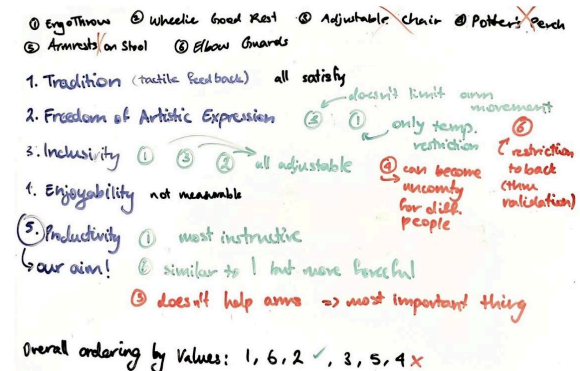


Figure 18. Converge based on the community's value.

In our Praxis II project, we used pairwise comparison as a structured tool to converge from a wide pool of posture-correcting concepts to a refined shortlist. While this tool is widely used among students, often defaulted to during convergence, we were intentional about using it with purpose, rather than simply following common practice. That mindset shaped how we approached the process and gave it specific meaning in the context of our project.

What set our use of pairwise comparison apart was how we integrated the MUD Clay Studio's community values into the decision-making process. Instead of comparing ideas solely based on vague preference or technical ease, we ranked and evaluated our concepts in relation to MUD's stated values: tradition, freedom of artistic expression, inclusivity, enjoyability, and productivity. This allowed us to customize the tool to reflect what mattered most to our users, not just what seemed easiest or most popular.

For each design concept, we discussed how well it aligned with each of these values. For example, concepts that limited arm mobility were ranked lower in terms of freedom of expression, while designs like the ErgoThrow scored higher due to their subtle, adjustable support that preserved artistic movement. By grounding our comparisons in stakeholder priorities, we made sure our convergence was value-driven, not just convenience-driven.

Using this tool helped us go from eight concepts to a shortlist of three. These were not just the most technically feasible ideas, they were the ones that aligned best with the posture goals and the values of the studio community.

I would still recommend pairwise comparison as a converging tool, because it's clear, scalable, and easy to use, but I offer this warning: it can lose its value when used automatically. Many students use it because it's what everyone else uses, without stopping to ask why. That's when it becomes dangerous, when it becomes a checkbox instead of a meaningful tool. The key is to always understand what you're evaluating, why you're evaluating it, and how to make the tool specific to your context.

Pairwise comparison helped us converge, but it only became valuable because we used it deliberately, to connect our ideas back to our framing and to the people we were designing for. The balance in engineering design is between big ideas and grounded logic, between creativity and clarity. It's not just about narrowing down options, it's about doing so in a way that truly serves people with purpose and meaning. And that, more than anything else, is what pushed me to grow.

CTMF #9: Functional Analysis

Purpose: To clearly explain what your design does, by breaking it down into its essential functions. It helps others understand how each part of the design contributes to solving the original problem

How do use it:

1. List the primary function(s)
 - What is the design fundamentally supposed to do?
2. Break it down into subfunctions
 - What smaller tasks must the design perform to achieve the main goal?
3. Link functions to features
 - Match each function to a specific part of your design

We used Functional Analysis in the representing stage to clearly explain why our design decisions mattered and how each concept fulfilled (or failed to fulfill) the core functions of a posture aid for beginner potters. At this point in the process, we weren't just justifying our final concept, we were using functional breakdowns to communicate how and why the ErgoThrow succeeded over alternatives like the Wheelie Good Rest.

This tool helped us refocus on the actual purpose of the device:

1. Encourage a natural forward lean
2. Promote elbow-to-torso contact
3. Provide something that reference designs don't already achieve

By comparing ErgoThrow and Wheelie against these specific functional goals, we could highlight that ErgoThrow created intentional physical contact (via Velcro) to guide posture, while Wheelie didn't actually introduce any new or adjustable contact. That difference wasn't just physical, it was functional, and that distinction allowed us to defend our concept more persuasively.

The most useful part of Functional Analysis is that it forces clarity. It made us explain not just what the product does, but why that function matters in context. For example, we realized Wheelie might seem

helpful but doesn't actually improve on the default posture issues. Meanwhile, ErgoThrow directly addressed beginner needs with minimal force and allowed posture development over time. It was through this structured analysis that we could confidently say, ErgoThrow > Wheelie.

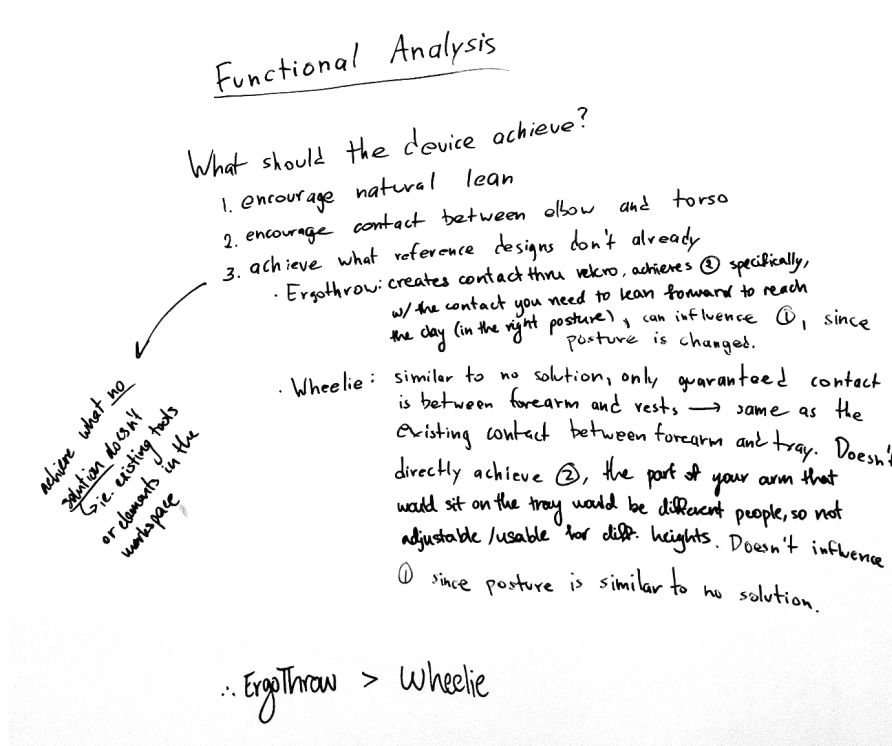


Figure 19. Functional analysis between ErgoThrow & Wheelie

Validation Results

- Back rest may not be optimal → for seasoned potters, the stool works well since there's so much movement, and they become experienced
- Don't add a backrest to ErgoThrow → ErgoThrow is for beginners, and back issues aren't beginner issues
- Lumbar support is good, anything to do to help back is good
- Back pain doesn't really factor in until you're less of a beginner
- As you get more advanced, body position changes
- Depending on cost, low cost for beginners is ok
- For seasoned potters, start to come away from the wheel, throw in all different positions, standing
- Potentially for production potters, mugs, repetitive strain
- Lifting belt maybe? Is it sufficient?
- This is very stationary, can get in the way of more mobile pottery work
- Guard is not as adjustable, maybe for physiotherapy applications?
- Consider separate between arm support and back brace → affect separate groups of potters

Would I use Functional Analysis again? Absolutely, but only when the core functions are well understood. I wouldn't recommend using it at the start of a project, because it can feel forced without a solid grasp on framing or user needs. When used at the right stage such as the presenting stage, it becomes a powerful way to demonstrate that your design decisions weren't just instinctive, but grounded in clear, deliberate reasoning.

Conclusion

Looking back on this portfolio, what stands out to me most isn't just the projects themselves, but how much my understanding of engineering design has evolved. When I first entered this program, I saw design as a way to chase big ideas, a tool to break free from limitations and make an impact. That belief hasn't changed, but it has become more grounded. I've come to understand that impactful design doesn't come from inspiration alone, it comes from intention, clarity, and the ability to use tools with purpose.

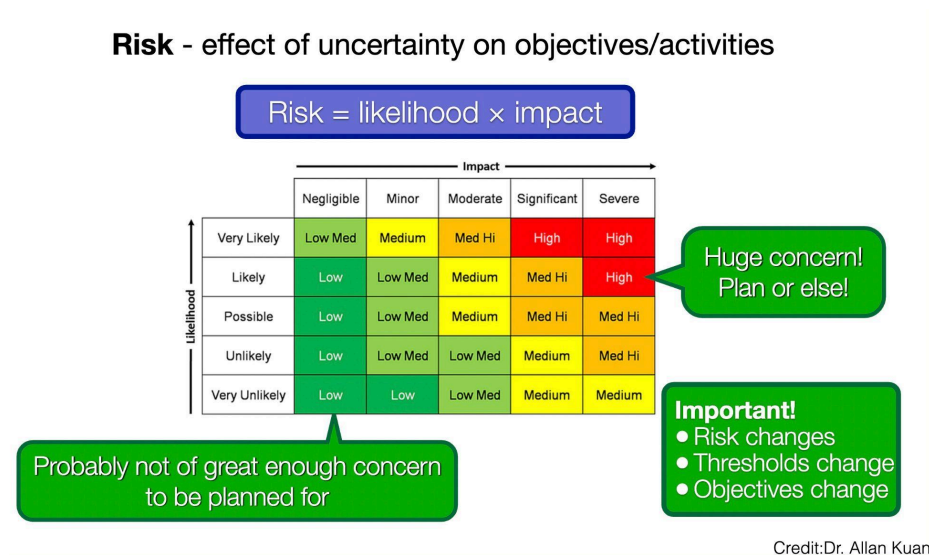
Throughout this year, I've learned to think more critically about each decision I make, not just what tool to use, but why I'm using it, what values it reflects, and how it connects to the people I'm designing for. More than anything, this portfolio represents my growth, not just in skill, but in mindset. I've learned that engineering isn't just about building things. It's about building them thoughtfully, responsibly, and meaningfully.

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- [1] C. Chen, H. Wu, I. Park, and A. Nasirov, *Creating Cat Litter Boxes for Better Cleanliness and Comfort – ESC101 Design Brief*, University of Toronto, 2024.
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Source Extracts

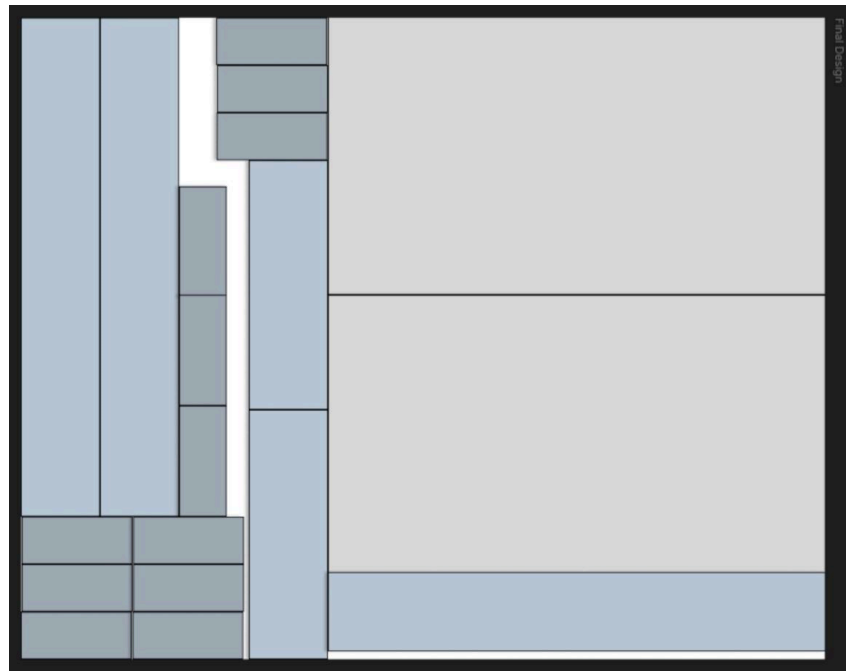
Extract A: Calculating Risk from Lecture Slides



Extract B: Risk Management Framework from Lecture Slides



Extract C: Bridge Dimension for CIV Project



Extract D: CIV Project Calculation

FOS against failure - design 0
 Sunday, November 24, 2024 12:46:50

Design 0:

From SFR and SFR:
 Maximum moment: $SFR = 6.9 \text{ kNm}$
 Maximum shear force: $V = 22.7 \text{ kN}$

Geometric properties:

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} = \frac{h_{20} w_{20} (w_{20} + \frac{h_{20}}{2}) + h_{20} w_{20} (\frac{w_{20}}{2}) + 2 h_{20} w_{20} (\frac{w_{20}}{2}) + 2 g_{20} (w_{20} - \frac{g_{20}}{2})}{h_{20} w_{20} + h_{20} w_{20} + 2 h_{20} w_{20} + 2 g_{20} w_{20}} = 41.43 \text{ mm}$$

$$I = \sum (I_{i0} + A_i \bar{y}_i^2) = \frac{w_{20} h_{20}^3}{12} + w_{20} h_{20} (\frac{w_{20}}{2} - \bar{y})^2 + \frac{w_{20} h_{20}^3}{12} + w_{20} h_{20} (\bar{y} - \frac{h_{20}}{2})^2 + 2 (\frac{w_{20} h_{20}^3}{12} + w_{20} h_{20} (\frac{w_{20}}{2} - \bar{y})^2) + 2 (\frac{g_{20} w_{20}^3}{12} + g_{20} w_{20} (\bar{y} - \frac{g_{20}}{2})^2) = 0.415 \cdot 10^8 \text{ mm}^4$$

$$Q_{max} = E (A_i \bar{y}_i^2) = (w_{20} h_{20}) (\bar{y} - \frac{h_{20}}{2})^2 + 2 (\bar{y} - w_{20}) (\bar{y})^2 = 43.50 \text{ mm}^3$$

$$Q_{she} = E (A_i \bar{y}_i^2) = h_{20} w_{20} (\bar{y} - \frac{h_{20}}{2})^2 = 2.210 \text{ mm}^3$$

Failure modes:

Case 1:
 $\sigma_{comp} = \frac{M(h-\bar{y})}{I} = 5.76 \text{ MPa}$
 $\sigma_{tension} = \frac{M \bar{y}}{I} = 6.86 \text{ MPa}$
 $\tau_{max} = \frac{Q_{max} V_{max}}{2 I b} = 10.5 \text{ MPa}$
 $\tau_{she} = \frac{Q_{she} V_{max}}{2 I b} = 0.108 \text{ MPa}$

Case 2:
 $\sigma_{comp} = \frac{0.85 \cdot 27 \text{ E}}{12 (1 - 0.4^2)} (\frac{h_{20}}{w_{20} - 2 g_{20}})^2 = 4.51 \text{ MPa}$

Case 3:
 $\sigma_{comp} = \frac{0.85 \cdot 27 \text{ E}}{12 (1 - 0.4^2)} (\frac{h_{20}}{w_{20} - 2 g_{20} / 2})^2 = 23.5 \text{ MPa}$

Case 4:
 $\sigma_{comp} = \frac{0.85 \cdot 27 \text{ E}}{12 (1 - 0.4^2)} (\frac{w_{20}}{w_{20}})^2 = 34.4 \text{ MPa}$

Case 5:
 $\tau_{max} = \frac{0.85 \cdot 27 \text{ E}}{12 (1 - 0.4^2)} (\frac{V_{max}}{w_{20}})^2 + (\frac{w_{20}}{w_{20}})^2 = 5.27 \text{ MPa}$

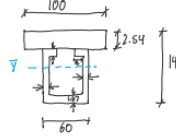
FOS:

FOS_{comp} = $\frac{\sigma_{allowable}}{\sigma_{actual}} = 4.28$
 FOS_{tension} = $\frac{\sigma_{allowable}}{\sigma_{actual}} = 1.08$
 FOS_{shear} = $\frac{\tau_{allowable}}{\tau_{actual}} = 3.36$
 FOS_{shear} = $\frac{\tau_{allowable}}{\tau_{actual}} = 1.84$
 FOS_{shear} = $\frac{\tau_{allowable}}{\tau_{actual}} = 5.01$
 FOS_{shear} = $\frac{\tau_{allowable}}{\tau_{actual}} = 0.992$

Conclusion: The bridge will fail due to bending at the top flange.

Extract E: CIV Project Calculation

FOS of Final Design Against Load Case 2
Monday, November 25, 2024 8:57 PM



From SFE and BME:
 $V_{max} = 304 \text{ N}$
 $N_{max} = 78.1 \times 10^3 \text{ N/m}$

Geometric Properties:

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} = \frac{h_{fl} w_{fl} (w_b + \frac{h_{fl}}{2}) + h_{fl} w_{fl} (\frac{w_b}{2}) + 2 h_{wb} w_{wb} (\frac{w_b}{2}) + 2 g_{wb} (w_b - \frac{g_{wb}}{2})}{h_{fl} w_{fl} + h_{fl} w_{fl} + 2 h_{wb} w_{wb} + 2 g_{wb}}$$

$$= 58.6 \text{ mm}$$

$$I = \sum (I_i + A_i \bar{y}_i^2) = \frac{w_{fl} h_{fl}^3}{12} + w_{fl} h_{fl} (w_b + \frac{h_{fl}}{2} - \bar{y})^2 + \frac{w_{fl} h_{fl}^3}{12} + w_{fl} h_{fl} (\bar{y} - \frac{h_{fl}}{2})^2 + 2 (\frac{h_{wb}^3}{12} + w_{wb} (\bar{y} - \frac{h_{wb}}{2})^2) + 2 (\frac{g_{wb}^3}{12} + g_{wb} (\bar{y} - \bar{y})^2)$$

$$= 1.930 \times 10^8 \text{ mm}^4$$

$$Q_{max} = \sum (A_i \bar{y}_i) = w_{fl} h_{fl} (\bar{y} - \frac{h_{fl}}{2}) + 2 \bar{y} w_{wb} (\bar{y}) = 8.90 \times 10^3 \text{ mm}^3$$

$$Q_{flue} = \sum (A_i \bar{y}_i) = w_{fl} h_{fl} (w_b + \frac{h_{fl}}{2} - \bar{y})^2 = 6.53 \times 10^3 \text{ mm}^3$$

Failure Modes:

$\sigma_{comp} = \frac{M(y - \bar{y})}{I} = 2.06 \text{ MPa}$
 $\sigma_{tens} = \frac{M \bar{y}}{I} = 3.59 \text{ MPa}$
 $\tau_{max} = \frac{Q_{max} V_{max}}{\sum I_b} = 0.552 \text{ MPa}$
 $\tau_{flue} = \frac{Q_{flue} V_{max}}{\sum I_b} = 0.06 \text{ MPa}$

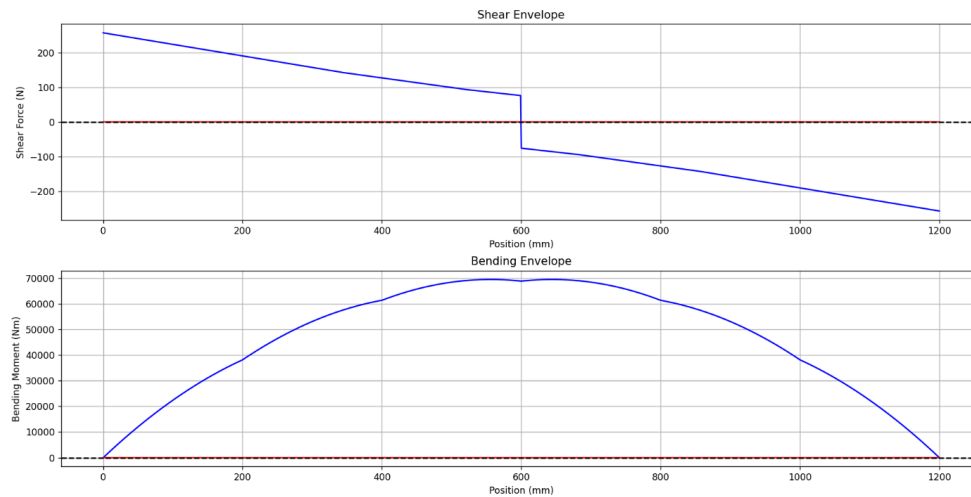
Thin Plate buckling:

Case 1:
 $\sigma_{comp} = \frac{H^2 E}{12(1-\nu^2)} \left(\frac{h_{fl}}{w_{fl}} \right)^2 = 6.84 \text{ MPa}$
 $\sigma_{comp} = \frac{0.1425 \pi^2 E}{12(1-\nu^2)} \left(\frac{h_{wb}}{(w_b - w_{wb})/2} \right)^2 = 23.49 \text{ MPa}$
 $\sigma_{comp} = \frac{6 \pi^2 E}{12(1-\nu^2)} \left(\frac{w_b - \bar{y}}{w_b} \right)^2 = 24.4 \text{ MPa}$
 $\tau_{tens} = \frac{5 \pi^2 E}{12(1+\nu)} \left[\left(\frac{w_b}{w_{fl}} \right)^2 + \left(\frac{w_b}{\bar{y}} \right)^2 \right] = 2.13 \text{ MPa}$

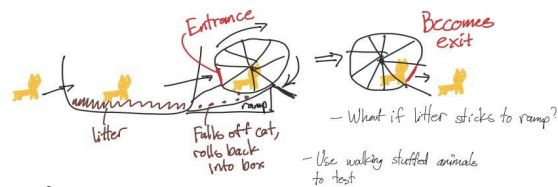
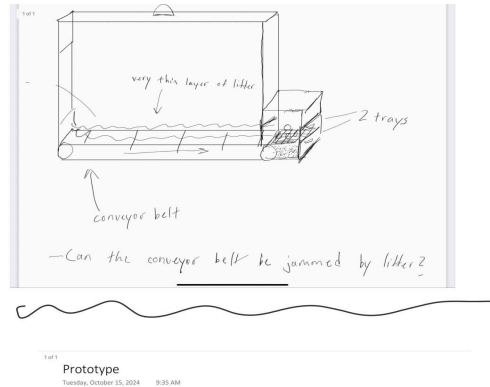
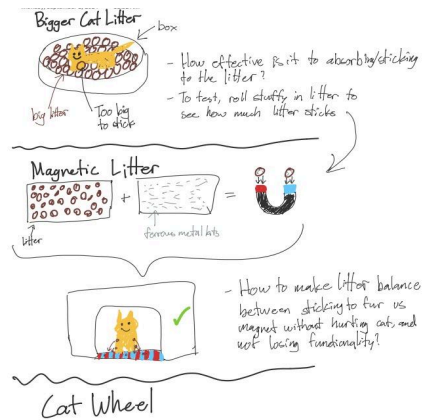
FOS's

$FOS_{comp} = \frac{\sigma_{compressive}}{\sigma_{comp}} = 2.96$
 $FOS_{tens} = \frac{\sigma_{tensile}}{\sigma_{tens}} = 8.37$
 $FOS_{shear} = \frac{\tau_{max, shear}}{\tau_{max}} = 7.25$
 $FOS_{flue} = \frac{\tau_{flue, shear}}{\tau_{flue}} = 24.4$
 $FOS_{buckling} = \frac{\tau_{buck}}{\tau_{max}} = 2.88$
 $FOS_{flexback} = \frac{\min(\sigma_{tensile})}{\sigma_{comp}} = 4.25$

Extract F: CIV Project SFD & BMD



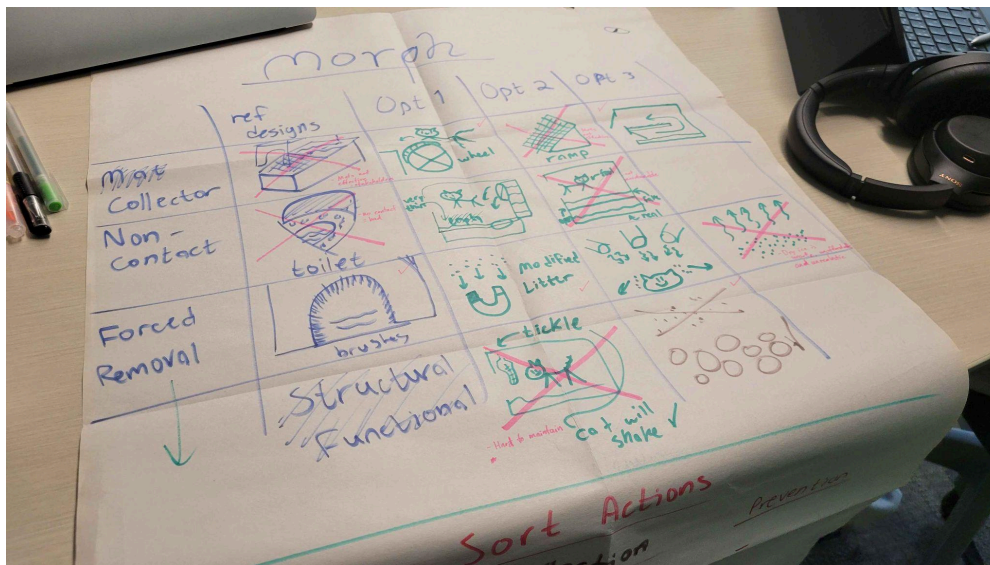
Extract G: Alpha Prototypes



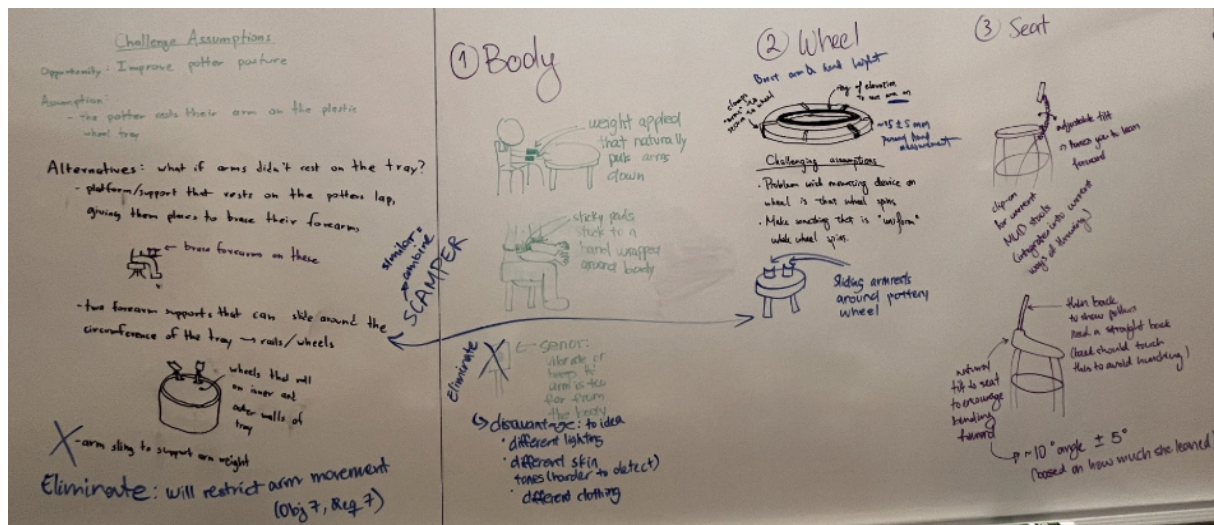
Extract H: Morph Chart example from Lecture Slides

Ways of achieving the Feature/function				
Features/Functions of what you're trying to design	Option 1	Option 2	Option 3	Option 4
	Vegetable picking device	Triangular plow	Tubular grabber	Mechanical picker
	Vegetable placing device	Rake	Rotating mover	Force from vegetable accumulation
	Dirt sifting device	Water from well	Sits in plow or carrier	
	Packaging device			
	Method of transportation	Track system	Sled	
Power source	Hand pushed	Horse drawn	Wind blown	Pedal driven

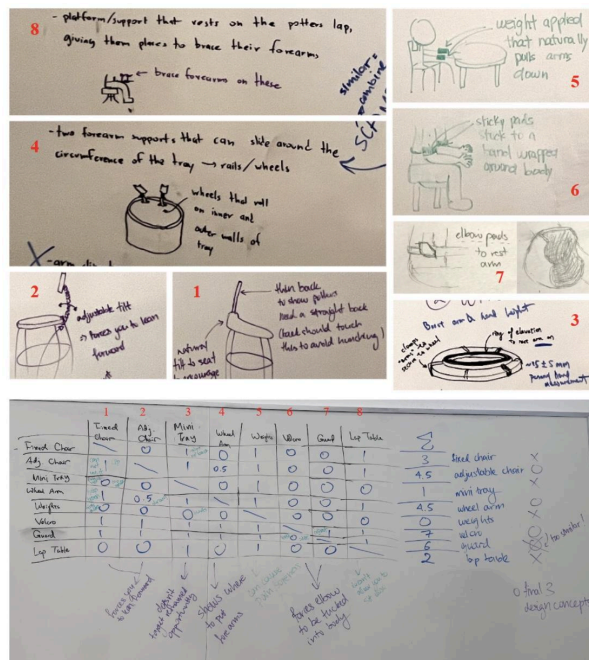
Extract I: Alpha Evidence of Morph Chart



Extract J: Rowe Award Diverging Process



Extract K: Showcase Evidence 1



Extract L: Showcase Evidence 2

Converging with Community's Values

- ① Ergonomic ② Wheelie Good Rest ③ Adjustable Chair ④ Pottery Perch
⑤ Armrest on Stool ⑥ Elbow Guards
1. Tradition (tactile feedback) all satisfy
2. Freedom of Artistic Expression
3. Inclusivity
4. Enjoyability not measurable
5. Productivity
6. our aim!
- ① most instructive
② similar to 1 but more forceful
③ doesn't help arms -> most important thing
- Overall ordering by Values: 1, 6, 2, 3, 5, 4

<https://mudtheclaystudio.com/pages/about>

The MUD Clay Studio has specific values which define their teaching culture and the environment in which our stakeholders (beginner potters) work in. MUD appreciates:

- Keeping the traditional aspect of pottery, ensuring there is tactile feedback for the potter with the clay and maintaining the same method of wheel throwing
- Allowing potters to express themselves artistically and explore their creative potential
- Creating an inclusive and diverse environment
- Nurturing passion and interest in pottery
- Having a productive and educational pottery journey

These were ordered using the information on MUD's website along with discussions with Hannun, the instructor.

Functional Analysis

What should the device achieve?

1. encourage natural lean
2. encourage contact between elbow and torso
3. achieve what reference designs don't already
 - Ergothrow: creates contact thru velcro, achieves ① specifically, w/ the contact you need to lean forward to reach the tray (in the right posture), can influence ①, since posture is changed.
 - Wheelie: similar to no solution, only guaranteed contact is between forearm and rests → same as the existing contact between forearm and tray. Doesn't directly achieve ②, the part of your arm that would sit on the tray would be different people, so not adjustable / usable for diff. heights. Doesn't influence ① since posture is similar to no solution.

achieve what no
solution doesn't
→ i.e. existing rests
or demands in the
workspace

∴ ErgoThrow > Wheelie